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Siamese Neural Networks for Wireless Positioning and Channel Charting

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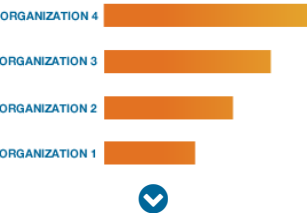
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Abstract: Neural networks have been proposed recently for positioning and channel charting of user equipments (UEs) in wireless systems. Both of these approaches process channel state information (CSI) that is acquired at a multi-antenna basestation in order to learn a function that maps CSI to location information. CSI-based positioning using deep neural networks requires a dataset that contains both CSI and associated location information. Channel charting (CC) only requires CSI information to extract relative position information. Since CC builds on dimensionality reduction, it can be implemented using autoencoders. In this paper, we propose a unified architecture based on Siamese networks that can be used for supervised UE positioning and unsupervised channel charting. In addition, our framework enables semisupervised positioning, where only a small set of location information is available during training. We use simulations to demonstrate that Siamese networks achieve similar or better performance than existing positioning and CC approaches with a single, unified neural network architecture.

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I. Introduction

Positioning of wireless transmitters under line-of-sight (LoS) propagation conditions is a relatively well understood topic [1] –[3]. Corresponding positioning techniques, such as triangulation and trilateration, find widespread use in global navigation satellite systems (GNSSs) and in localization systems that leverage user equipment (UE) access to multiple infrastructure basestations (BSs). However, positioning of wireless transmitters is known to be significantly more challenging under complex propagation conditions [4], [5]. Such conditions appear, for example, in non-LoS scenarios (which may happen indoors) or for channels with rich multi-path components (which may happen in dense urban areas). The trend towards communication at high carrier frequencies, such as millimeter-wave systems, further increases the prevalence of complex propagation conditions as channel properties can abruptly change in space [6], [7].

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